

Francesco Deangelis

Curriculum vitae

Münster, NRW, Germany

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🌐 francescodeangelis.github.io

Experience

- 10/2024–present **Postdoctoral Researcher in Mathematics**, *University of Münster*, Germany
Mentor: Prof. Caterina Ida Zeppieri
- 10/2023–11/2023 **Short-Term Visitor**, *Institute for Advanced Study*, Princeton, NJ, USA
Visiting Prof. Camillo De Lellis (1 October – 30 November 2023)

Education

- 11/2020–7/2024 **PhD in Mathematics**, *Gran Sasso Science Institute*, L'Aquila, Italy
cum laude
Advisor: Prof. Camillo De Lellis (IAS, Princeton)
Thesis: Boundary regularity for the Mumford–Shah functional with Dirichlet boundary conditions
- 10/2017–3/2020 **MSc in Pure and Applied Mathematics**, *University of Rome Tor Vergata*, Italy
110/110 cum laude
Advisor: Prof. Michiel Bertsch
Thesis: A free boundary problem arising in the theory of binary fluids
Award: €2000 prize for the master's thesis
- 10/2014–7/2017 **BEng in Mechanical Engineering**, *University of Rome Tor Vergata*, Italy
110/110 cum laude
Advisor: Prof. Giovanni Bellettini
Thesis: A variational approach to fracture evolution
- 2013–2014 **BSc in Economics and Industrial Organization**, *University of Warwick*, UK
Left before completion in April 2014 to pursue studies more aligned with my academic interests
- 2008–2013 **Diploma di Maturità Classica**, *Liceo Classico Ugo Foscolo*, Albano Laziale (Rome), Italy
100/100 cum laude

Publications

- 1) Roberto Benzi, Michiel Bertsch, and Francesco Deangelis. A free boundary problem for binary fluids. *Interfaces Free Bound.*, 23(4):485–506, 2021. URL: <https://ems.press/journals/ifb/articles/3324283>, doi:10.4171/IFB/462.
- 2) Francesco Deangelis. Boundary regularity for the Mumford-Shah functional with Dirichlet boundary conditions, 2024. [arXiv:2404.07162](https://arxiv.org/abs/2404.07162).
- 3) Francesco Deangelis, Matteo Focardi, and Caterina Ida Zeppieri. Stochastic homogenization of fractional obstacle problems, 2026. [arXiv:2604.09896](https://arxiv.org/abs/2604.09896).

- 4) Francesco Deangelis. Solution of the Mumford–Shah conjecture, 2026. arXiv: 2609.26732.

Talks

- 01/07/2026 **Stochastic homogenization of fractional obstacle problems**, *Oberseminar Stochastik, University of Münster, Germany*
- 11/02/2026 **Boundary regularity for the Mumford–Shah functional with Dirichlet boundary conditions**, *XXXV Convegno Nazionale di Calcolo delle Variazioni, Riccione, Italy*
- 30/01/2026 **Stochastic homogenization of fractional obstacle problems**, *Analysis Seminar, University of Florence, Italy*
- 03/12/2025 **Boundary regularity for the Mumford–Shah functional with Dirichlet boundary conditions**, *Analysis Seminar, University of Parma, Italy*
- 26/11/2025 **Stochastic homogenization of fractional obstacle problems**, *CASA Colloquium, Eindhoven University of Technology, Netherlands*
- 29/10/2025 **An introduction to stochastic homogenization**, *University of Münster, Germany*
- 10/04/2025 **Homogenization on randomly perforated domains**, *Postdoc Colloquium, University of Münster, Germany*
- 03/07/2024 **Boundary regularity for the Mumford–Shah functional with Dirichlet boundary conditions**, *Analysis Seminar, Gran Sasso Science Institute, L'Aquila, Italy*

Teaching

- 2025 **Exercise Class Instructor - Applied Functional Analysis**, *University of Münster*
- 2018-2019 **Tutor - Mathematical Analysis 1**, *University of Rome Tor Vergata*

Languages

- Italian Native
- English Fluent — TOEFL iBT 115/120

Computer skills

- Web programming PHP, MySQL, AJAX, CodeIgniter, jQuery, JavaScript, HTML, CSS, Bootstrap, Python, Django, Jekyll
- Mobile programming Flutter
- Scientific programming MATLAB, Python, C